

THE AXIS TEAM

HFES Regional Meeting

The Tufts University Idea Lab · Spring 2026

Electrical Engineering

Mechanical Engineering

Human Factors Engineering

Today's Presentation

01

Project Overview

Problem statement & goals

02

Project Background & Prototype

First Prototype, our improvements since

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Electronics

Circuit design, power systems & firmware

04

Human Factors Engineering

REBA/RULA analysis, task analysis & user interviews

05

Future Plans

Roadmap, milestones & open questions

The Axis Team

Problem Statement

Guitar pedals are essential, but they're fundamentally linear. Most musicians can only manipulate one effect at a time, via footswitch or dial. That paradigm limits the performer's expressive control and disrupts the flow of live creativity.

Project Goals

Develop a motion-based control system that allows guitarists to manipulate sound in real time through natural movement. The design should be lightweight, unobtrusive, and easy to use, while supporting intuitive, expressive control.

Team Composition

EE Team

Leads: Paul, Nahom

ME Team

Leads: Ryan, Seamas

HFE Team

Leads: Rianna

Project Background

First Prototype

- Our first ever design used a large C-clamp mounted at the base of the guitar with a breadboard and exposed electronics.
- We then shifted to a more user-centered design, moving the system to the headstock where motion is more expressive.



Project Background

Improvements

- Snark Tuner
- Headstock Housing



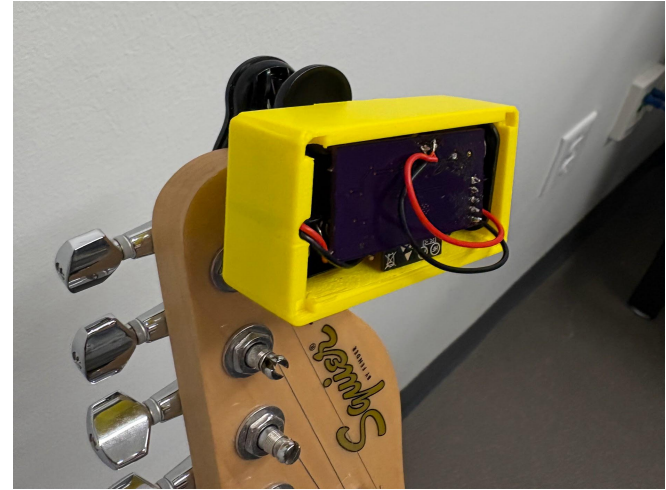
Project Background

Improvements

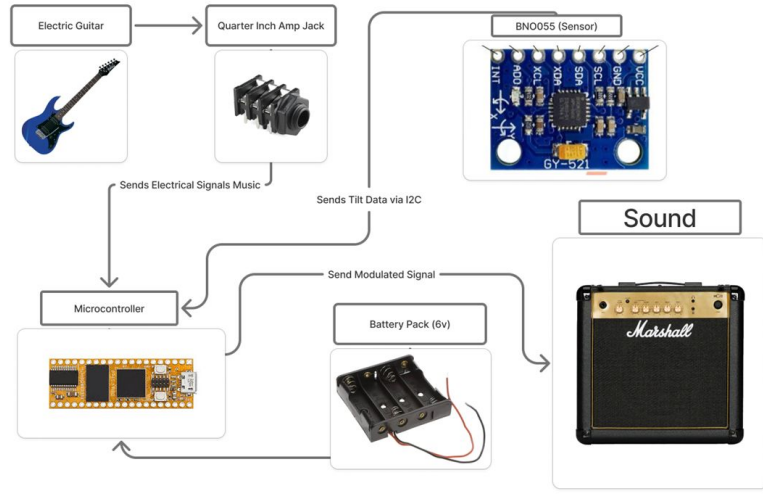
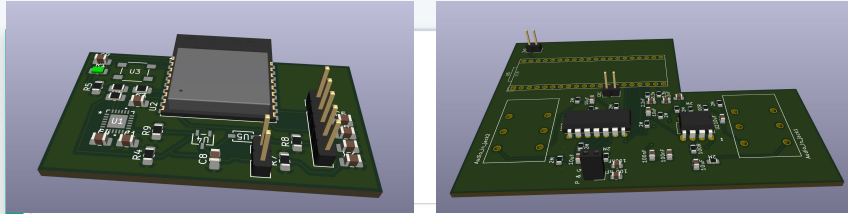
- Stabilizing Fins
- Moved Ball Joint
- Battery Integration



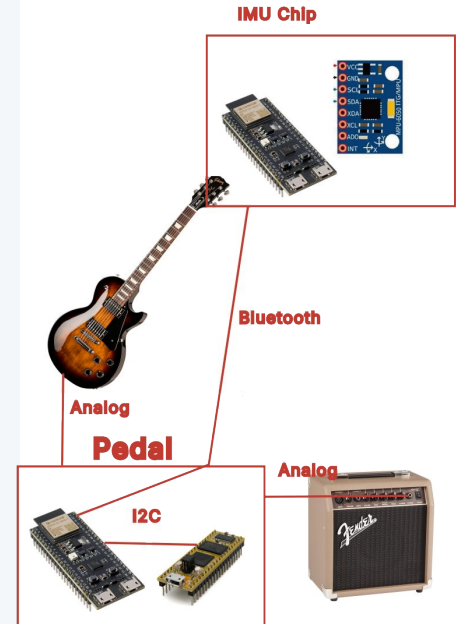
Current Prototype



PCB Design & Functional Overview



- Small lightweight PCB
- External audio components to manage sound
- BLE for fast angle readings



Goals	REBA / RULA	Task Analysis	User Interviews
	Identify current challenges and pain points while playing	- Observe current habits - Analyze impact of playing posture on pedal use & movement	- Identify usability issues - Understand user likes & dislikes of prototype - Assess intuitiveness
Analysis	- Understood the pain points of playing a guitar - Generated ongoing user needs list	- A lot of movements are exaggerated when performing - There is a lot of walking across stage	- Lid may fall off during performance - Familiar guitar attachments easier to use

Future Plans



- Explore app or LCD interface for effect control
- Develop fully integrated, compact housing with internal battery
- Improve usability (on/off button, system status LED)
- Conduct real-world testing with guitarists to refine design

Thank You

Questions & Discussion

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